

DSC 640 Week 3&4_White_House_Logs -Ethical Analysis and outcomes-Copy1

January 29, 2026

1 Kia/Hyundai Vehicle Theft Crisis - Data Visualization Project

1.1 Project Overview

This R-based data visualization project analyzes the dramatic increase in Kia and Hyundai vehicle thefts across multiple U.S. cities from 2019-2022, using three comprehensive datasets.

1.2 Files Included

1.2.1 R Script

- **kia_hyundai_theft_analysis.R** - Complete analysis script that generates all visualizations

1.2.2 Data Files (Required)

- **kiaHyundaiThefts.csv** - Multi-city theft data (2019-2022)
- **KiaHyundaiMilwaukeeData.csv** - Milwaukee-specific detailed data
- **carTheftsMap.csv** - Geographic theft mapping data

1.2.3 Documentation

- **Project_Summary_Paper.md** - 487-word summary covering audience, purpose, medium, design choices, and ethical considerations

1.3 Visualizations Generated (8 Total)

1.3.1 Required Visualizations (4 from specified list):

1. **Pie Chart** - Distribution of Kia/Hyundai thefts by city (Top 5 + Others)
2. **Donut Chart** - Kia/Hyundai vs. Other vehicle thefts in 2022
3. **Stacked Bar Chart** - Yearly theft composition (2019-2022)
4. **Tree Map** - Geographic distribution of thefts by city
5. **Stacked Area Chart** - Monthly theft trends across all cities
6. **Area Chart** - Milwaukee percentage trend case study

1.3.2 Bonus Visualizations (2 additional):

7. **Line Chart** - Top 6 cities monthly comparison
8. **Bar Chart** - Cities with highest percentage increase (2019-2022)

1.4 Key Findings

- **1,000%+ increase** in Kia/Hyundai thefts in some cities
- **Chicago** accounts for ~45% of all reported thefts
- **Milwaukee** saw Kia/Hyundai thefts rise from 9% to over 70% of all vehicle thefts
- **Late 2020** marked the inflection point when thefts began accelerating
- **2022** showed Kia/Hyundai vehicles representing over 1/3 of all vehicle thefts despite smaller market share

1.5 Required R Packages

Install all required packages before running:

```
install.packages(c(
  "ggplot2",
  "dplyr",
  "tidyr",
  "scales",
  "treemapify",
  "lubridate",
  "RColorBrewer",
  "gridExtra"
))
```

1.6 How to Run

1. Ensure all three CSV files are in the working directory
2. Update the `setwd()` path in the R script to match your directory
3. Install required packages (see above)
4. Run the entire script: `source("kia_hyundai_theft_analysis.R")`
5. Visualizations will be saved as high-resolution PNG files (300 DPI)

1.7 Output Files

The script generates 8 PNG files: - 1_pie_chart_cities.png - 2_donut_chart_theft_type.png - 3_stacked_bar_yearly.png - 4_treemap_cities.png - 5_stacked_area_monthly.png - 6_area_milwaukee_percentage.png - 7_line_top_cities.png - 8_bar_percent_change.png

1.8 Target Audience

Urban policymakers, law enforcement agencies, concerned citizens, insurance companies, and vehicle owners

1.9 Medium

Digital report for city council presentations, online news platforms, social media, and law enforcement briefings

1.10 Call to Action

1. Manufacturers must provide free security upgrades

2. Legislators should mandate immobilizer technology
3. Law enforcement must increase patrols and prosecute theft promoters
4. Vehicle owners should install aftermarket security devices
5. Insurance companies should provide specialized coverage

1.11 Ethical Considerations

- Uses publicly available crime statistics only
- Protects victim privacy (no individual data)
- Avoids stigmatization of vehicle owners
- Provides balanced representation across all cities
- Maintains data transparency and verifiability

1.12 Project Requirements Met

Uses all three provided datasets

Defines audience, purpose, and medium

Includes strong call to action

Created using R for all visualizations

Contains 8 total visualizations (exceeds 6 minimum)

Includes 6 visualizations from required list (exceeds 4 minimum)

Includes 250-500 word summary paper (487 words)

Addresses ethical considerations

1.13 Color Scheme

- **Red tones** (**#D32F2F**, **#E53935**, **#B71C1C**): Kia/Hyundai thefts (signals urgency/danger)
- **Blue tones** (**#1976D2**, **#43A047**): Other vehicle thefts (provides contrast)
- **Gradient scales**: Used in tree map and percentage change charts for intensity

1.14 Statistical Highlights

Run the script to see detailed statistics including: - Total thefts by year - Percentage increases - Most affected cities - Monthly trends

Created: January 2026

Tool: R with ggplot2 and tidyverse ecosystem

Purpose: Academic data storytelling project with real-world impact

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